



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

EXPERIMENT 6

Aim: Data Anonymization Using Masking and Redaction Techniques in Data Engineering.

Theory:

- Implementing data anonymization using masking and redaction techniques is crucial for protecting sensitive information in data engineering. Here's a step-by-step guide to help you get started:
- Understand the Techniques
 - Data Masking: This involves replacing sensitive data with fictional but realistic data. For example, replacing a real credit card number with a fake one that has the same format.
 - Data Redaction: This involves permanently removing or obscuring sensitive information. For example, blacking out parts of a document.
- Identify Sensitive Data
- Determine which data needs to be anonymized. This typically includes personally identifiable information (PII), financial data, and health records.
- Choose the Right Tools
- Use tools and libraries that support data masking and redaction. Some popular options include:
 - Apache NiFi: For data flow automation and transformation.
 - Talend: For data integration and anonymization.
 - PII Tools: For de-identification, masking, and redaction (<https://pii-tools.com/pii-de-identification-vs-masking-vs-redaction/>).
- Python API

Implement Data Masking

- Static Data Masking: Apply masking to data at rest. This is useful for non-production environments like testing and development.
- Dynamic Data Masking: Apply masking on-the-fly to data in use. This is useful for production environments where data needs to be accessed but not exposed.
- Implement Data Redaction
- Manual Redaction: Manually redact sensitive information in documents and files. This is often used for legal and compliance purposes.



Name: Gargi Sukhatankar

SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

- Automated Redaction: Use software to automatically detect and redact sensitive information in large datasets and documents (<https://bigid.com/blog/data-redaction/>).
-
- Test and Validate
- Ensure that the anonymized data retains its usability for analysis and processing.
- Validate that the masking and redaction processes effectively protect sensitive information.
- Monitor and Maintain
- Regularly review and update your anonymization processes to adapt to new types of sensitive data and regulatory requirements.
- By following above steps, you can effectively implement data anonymization using masking and redaction techniques in your data engineering projects. If you have any specific questions or need further details, feel free to ask! escalations.

OUTPUT:

Step 1: Install Required Libraries

```
[42] import pandas as pd
import re
from faker import Faker
```

Step 2: Load the CSV

```
[41] import pandas as pd

df = pd.read_csv("/content/employee_info.csv")
df.head()
```

	Name	Email	Phone
0	Tiya	tiya@gmail.com	9876543210
1	Eevee	evee@example.com	9123456789
2	Ash	ash@poke.com	9988776655

Step 3: Mask the Data using Faker



Name: Gargi Sukhatankar

SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

```
from faker import Faker
fake = Faker()

df['Masked_Name'] = df['Name'].apply(lambda x: fake.name())
df['Masked_Email'] = df['Email'].apply(lambda x: fake.email())
df['Masked_Phone'] = df['Phone'].apply(lambda x: fake.numerify('##*****##'))
```

Step 4: Redact the Data

```
df['Redacted_Name'] = '[REDACTED]'
df['Redacted_Email'] = '[REDACTED]'
df['Redacted_Phone'] = '[REDACTED]'
```

Step 5: Display table with masked and redacted rows

```
print("Original Data:")
print(df[['Name', 'Email', 'Phone']])
print("\n")

print("Masked Data:")
print(df[['Masked_Name', 'Masked_Email', 'Masked_Phone']])
print("\n")

print("Redacted Data:")
print(df[['Name', 'Redacted_Email', 'Redacted_Phone']])
```



Name: Gargi Sukhatankar

SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

```
➡ Original Data:
      Name      Email      Phone
0   Tiya   tiya@gmail.com  9876543210
1  Eevee  evee@example.com  9123456789
2    Ash   ash@poke.com   9988776655

Masked Data:
      Masked_Name      Masked_Email  Masked_Phone
0   Daniel Lopez  thomascynthia@example.org  31*****70
1     Tina Roach  alexandrahood@example.org  85*****59
2  Andrew Thompson  leah66@example.net  96*****23

Redacted Data:
      Name  Redacted_Email  Redacted_Phone
0   Tiya      *****      *****
1  Eevee      *****      *****
2    Ash      *****      *****
```

Conclusion:

In this experiment, we successfully demonstrated the concepts of data masking and data redaction using a sample dataset containing sensitive information such as names, emails, and phone numbers. We used the pandas library to handle CSV files and the Faker library to generate realistic fake data for masking purposes.

Masking helped in hiding the original data while preserving its format and usability, making it ideal for testing and development without compromising privacy. On the other hand, redaction replaced sensitive data with placeholders like ***** ensuring complete removal of confidential information from the dataset.

This approach is crucial in real-world applications where data privacy and security are essential, especially when handling personally identifiable information (PII). Through this experiment, we gained hands-on experience with Python libraries for privacy-preserving data processing.